R Energy Project: Session Notes

DATA 607 Project 2 (Dataset 2): electricity production by country, Oct 11-12, 2026

A reference of each question worked through in this session, the fix, and the code. Dataset: *electricity_production_by_country.csv* (216 rows; TWh by source for each country, reporting years mostly 2024-2025).

1. Removing unwanted characters from a string column

Use **gsub()** (base R) or **str_remove_all()** (stringr). Square brackets list characters to remove; **^** inside brackets means "anything except". Escape regex specials (. ( ) * + ? |) with \\ or use fixed = TRUE.

```
df$col <- str_remove_all(df$col, "[$,]")       # remove every $ and ,
df$col <- str_remove(df$col, "_x$")            # only at the end
df$col <- str_remove_all(df$col, "[^0-9.]")    # keep digits and decimals
```

2. Doing it inside a pipeline

```
df_clean <- df |>
  mutate(price = str_remove_all(price, "[$,]") |> as.numeric(),
         name  = str_trim(name))

# same treatment across several columns
df |> mutate(across(c(price, cost), ~ str_remove_all(.x, "[$,]") |> as.numeric()))
```

3. Adding population data

Built **country_population.csv** (Location, Year, Population) from World Bank total population (SP.POP.TOTL), 2009-2025, with country names renamed to match the energy file (e.g. "Korea, Rep." to South Korea, "Turkiye" to Turkey). 203 of 216 rows match; the 13 gaps are places the World Bank doesn't report separately (Taiwan, French overseas departments, small territories).

4. Joining on two columns

Joining on both Location and Year gives each country the population for its own reporting year.

```
elec_pop <- elec |> left_join(pop, by = c("Location", "Year"))
elec_pop <- elec |> left_join(pop, join_by(Location, Year))          # dplyr 1.1+
elec_pop <- elec |> left_join(pop, by = c("Location" = "Country"))   # different names
```

If the join errors, check that Year has the same type in both tables (as.integer()).

5. Turning off scientific notation

```
options(scipen = 999)                                  # whole session
format(8.2e9, scientific = FALSE, big.mark = ",")      # one value, as text
scales::comma(x)                                       # tables and axis labels
options(pillar.sigfig = 10)                            # tibble printing
```

6. Unit conversions from TWh

Multiply TWh by	Result
1e3	GWh
1e6	MWh
1e9	kWh (use for per-capita)
1e12	Wh

Multiplying by 1e7 has no SI prefix (it yields units of 100 kWh).

```
mutate(Per_Capita_Energy_kWh = Value_TWh / Population * 1e9)
```

7. Empty scatter plot: column name typo

The column was created as **Per_Capita_Energy_kWh** but plotted as **Per_Capita_energy_kWh**. R is case-sensitive. If a plot is still blank, check nrow() and that the filter value (e.g. "Total") matches exactly. Log scales help because China and India dwarf other countries; drop the World row.

```
ggplot(Energy3, aes(x = Population, y = Per_Capita_Energy_kWh)) +
  geom_point() +
  scale_x_log10(labels = scales::comma) + scale_y_log10(labels = scales::comma)
```

8. Highlighting outliers

Flag outliers in a new column, map color to it, and label them with ggrepel. Either take the top N, or use the 1.5 x IQR rule on the log scale (raw values are too skewed).

```
Energy3 <- Energy3 |>
  mutate(log_kwh = log10(Per_Capita_Energy_kWh),
         Outlier = log_kwh > quantile(log_kwh, .75) + 1.5 * IQR(log_kwh) |
                   log_kwh < quantile(log_kwh, .25) - 1.5 * IQR(log_kwh))

ggplot(Energy3, aes(Population, Per_Capita_Energy_kWh, color = Outlier)) +
  geom_point() +
  ggrepel::geom_text_repel(data = filter(Energy3, Outlier),
                           aes(label = Location), size = 3, show.legend = FALSE) +
  scale_color_manual(values = c("FALSE" = "grey60", "TRUE" = "red"))
```

9. Charts for the energy-source mix

World mix: a sorted horizontal bar chart of shares (clearer than a pie with nine slices). **Across countries:** 100% stacked bars via geom_col(position = "fill") for the top producers.

10. Classifying sources as Clean or Dirty

A named vector works as a lookup table. Sources not in it (the "Total" rows) become NA.

```
class_energy <- c(Coal = "Dirty", Gas = "Dirty", Other_fossil = "Dirty", Bioenergy = "Dirty",
                  Nuclear = "Clean", Hydro = "Clean", Wind = "Clean", Solar = "Clean",
                  Other_renewables = "Clean")

Energy2 <- Energy2 |> mutate(Energy_Class = unname(class_energy[Energy_Source]))
```

Alternative: dplyr::case_match(). Options for showing source and class together: a stacked bar per class, bars per source colored by class, facet_wrap(~ Energy_Class), or a treemap (treemapify).

11. One bar showing both source and class

Use a constant x value so everything stacks on one bar, order the factor levels so each class sits together, and use green shades for clean and brown shades for dirty sources.

```
source_order <- c("Coal", "Gas", "Other_fossil", "Bioenergy",
                  "Nuclear", "Hydro", "Wind", "Solar", "Other_renewables")

Energy4 |>
  mutate(Energy_Source = factor(Energy_Source, levels = rev(source_order))) |>
  ggplot(aes(x = "World", y = Total_Energy_Use, fill = Energy_Source)) +
  geom_col(width = 0.5) +
  scale_fill_manual(values = source_cols, breaks = source_order) +
  scale_y_continuous(labels = scales::comma) +
  coord_flip() +
  labs(x = NULL, y = "TWh", fill = "Source",
       subtitle = paste(class_totals$label, collapse = "   |   "))
```

12. Quarto render error: could not find function "+<-"

subtitle = was written outside labs() as + *subtitle* = ..., so R read the plot chain as the target of an assignment. Fix: move subtitle inside labs() (as above), and make sure class_totals is created before the plot.